PREFACE

# Rare Bleeding Disorders

This issue of *Seminars in Thrombosis and Hemostasis* is devoted to the topic of rare bleeding disorders (RBDs) caused by coagulation factor defects and deficiencies. The issue begins with an introductory article by the guest editor and some colleagues that primarily looks at the incidence of the separate RBDs as well as some of their features. The remaining articles within this issue have been prepared by experts in the field, each of who were asked to focus on individual or specific cases of RBDs. Specifically, each of the authors were asked to detail, for each RBD, the clinical manifestations, the laboratory assays used in the diagnosis of the specific RBD, the phenotype and genotypic analysis, problems with the laboratory evaluation, prenatal diagnosis (if required), and treatment options, with an additional focus on treatment of women suffering these rare disorders.

The prevalence of RBDs in the general population is low, and homozygous or double heterozygous deficiencies vary from 1:500,000 for factor VII deficiency to 1:2,000,000 for prothrombin and FXIII deficiencies (Table 1). This low prevalence leads to significant diagnostic challenges, because the likelihood of identifying such disorders is in general low but at the same time certainly not improbable. Thus, although these disorders are generally not foremost in the minds of treating clinicians and testing laboratories, they can occur, and their possibility should always be considered by clinicians and laboratory scientists alike after discounting other more common disorders or in those cases where familial history might otherwise be suggestive. Indeed, it is likely that many of us will actually come across at least one of each of these disorders over the period of our careers.

At the same time, the rarity of these disorders means that definitive information is lacking about presenting symptoms, the levels of specific factor(s) required for supporting normal hemostasis, and also for surgical hemostasis, as are specific guidelines on their diagnosis and treatment. That the appropriate diagnosis, monitoring, and treatment of affected individuals will be compromised by this paucity of knowledge and the limited understanding among general care centers was the main driving force behind the preparation of this issue of *Seminars in Thrombosis and Hemostasis.*

We expect that readers of this journal will find this issue to be not only of considerable interest but also a valuable long-term future reference source for both clinical and laboratory practice. In addition, a summary of some of the major observations for the RBDs is provided in Table 1.

We thank all the contributors to this issue for their thoughtfully considered articles, and we also thank Roberta Palla, Marzia Menegatti, and Pier Mannuccio Mannucci for their valuable assistance in the preparation of this issue. We are grateful also to Giuseppe Lippi for undertaking an additional review of the content of each article prior to publication.

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## REFERENCE

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**Table 1 Rare Bleeding Disorders: A Comparative Summary***

| Rare Bleeding Disorder | Afibrinogenemia; Hypofibrinogenemia; Dysfibrinogenemia | Prothrombin Deficiency | Factor V Deficiency | Combined Factor V and Factor VIII Deficiency | Factor VII Deficiency | Factor X Deficiency | Factor XI Deficiency | Factor XIII Deficiency | Multiple Vitamin K–Dependent Factor Deficiency |
|---|---|---|---|---|---|---|---|---|---|
| **Gene(s) responsible** | *Fibrinogen Bβ (FGB)*<br>*Fibrinogen Aα (FGA)*<br>*Fibrinogen γ (FGG)* | *Prothrombin (F2)* | *Factor V (F5)* | Lectin mannose binding protein (*LMAN1 or ERGIC-53)* and multiple coagulation factor deficiency 2 (*MCFD2)* | *Factor VII (F7)* | *Factor X (F10)* | *Factor XI (F11)* | *Factor XIIIA (F13A)* *factor XIIIB (F13B)* | γ-Glutamyl carboxylase (GGCX) and vitamin K epoxide reductase (VKORC1) |
| **Approximate prevalence of RBDs** | **Afibrinogenemia:** 1:1,000,000; **hypofibrinogenemia** and **dysfibrinogenemia** more common | **Aprothrombinemia:** incompatible with life. **Prothrombin deficiency:** 1:2,000,000. | 1:1,000.000 | 1:1,000.000 | 1:500,000 | 1:1,000,000 | 1:1,000,000 | 1:2,000,000 | Not known |
| **Frequency among RBDs**[†] (*see Ref. 1*) | 8% to 10% | 2% | 7% to 9% | 3% to 15% | 21% to 34% | 8% to 10% | 23% to 39% | 5% to 7% | Not known |
| **Factor(s) affected (common name[s]/ alternate name[s])** | Fibrinogen/factor I | Prothrombin/factor II | Factor V | Factor V and Factor VIII | Factor VII | Factor X | Factor XI | Factor XIII | FII/FVII/FIX/FX |
| **Major function(s) of factor(s)** | Major clotting protein; converts to fibrin | On conversion to thrombin promotes fibrin clot formation; displays anticoagulant activities through activation of the thrombomodulin/protein C pathway | FVa assembles with FXa on phospholipid membranes, forming a complex that enhances the rate of prothrombin activation. FV also contributes to the anticoagulant pathway by downregulating FVIII activity. | The LMAN1-MCFD2 protein complex functions as a cargo receptor that facilitates the transport of FV and FVIII from the ER to the Golgi apparatus | FVIIa, complexed with tissue factor, generates a burst of activated FIX and FX ultimately leading to formation of a stable fibrin clot | Complexed with FVa, FXa accelerates to 280,000-fold thrombin formation. FXa activates also FV, FVIII, and FVII. | FXIa acts by cleaving FIX in the intrinsic blood-coagulation pathway. It also displays antifibrinolytic properties by promoting activation of thrombin activatable fibrinolysis inhibitor. | FXIIIa forms fibrin γ-chain dimers, to cross-link its α-chains into high-molecular-weight polymers, and to attach α2 plasmin inhibitor to fibrin α-chains | Hepatic γ-glutamyl carboxylase (GGCX) catalyzes the γ-carboxylation of vitamin K–dependent coagulation factors, requiring reduced vitamin K (KH2) as a cofactor. During the γ-carboxylation reaction, KH2 is converted to vitamin K epoxide (KO), which is recycled to KH2 by the vitamin K VKORC1 |
| **Site(s) of synthesis** | Liver | Liver | Liver and megakaryocytes | **LMAN1:** highest expression in skeletal muscle, kidney, liver, and placenta. **MCFD2:** expressed in multiple tissues. | Liver | Liver | Liver and megakaryocytes | **FXIIIA**: cells of bone marrow origin. **FXIIIB**: liver. | Liver |
| **Approximate biological half-life** | 2–4 days | 70 hours | 36 hours | **FV**: 36 hours. **FVIII**: 10–14 hours. | 4–6 hours | 40–60 hours | 50 hours | 9–12 days | See corresponding factors |
| **Typical normal reference range**[‡] | 1.5–3.5 g/L | 75% to 113% | 70% to 134% | FV: 70% to 134%. FVIII: 51% to 147%. | 62% to 138% | 66% to 126% | 70% to 130% | 53% to 221% | See corresponding factors |
| **Main bleeding sites and/or symptoms for severe deficiencies** | **Afibrinogenemia**: Common:<br>- Umbilical cord bleeding | Common:<br>- Hematomas<br>- Bruising<br>- Hemarthrosis<br>- After tooth extraction | Common:<br>- Epistaxis<br>- Menorrhagia<br>- Oral cavity | Common:<br>- Easy bruising<br>- Epistaxis<br>- Gum bleeding<br>- After surgery<br>- After dental extraction | Common:<br>- Easy bruising<br>- Epistaxis<br>- Gum bleeding<br>- Menorrhagia<br>- After surgery | Common:<br>- Epistaxis<br>- Menorrhagia<br>- Hemarthrosis<br>- Hematomas<br>- Posttrauma | Most bleeding manifestations are injury-related. Women are prone to excessive bleeding during menstruation. | Common:<br>- Umbilical cord bleeding<br>- Bruising<br>- Subcutaneous hematoma | - Intracranial (at birth)<br>- Umbilical stump<br>- Hemarthrosis<br>- Retroperitoneal soft tissue |

**Table 1** (*Continued*)

| Rare Bleeding Disorder | Afibrinogenemia; Hypofibrinogenemia; Dysfibrinogenemia | Prothrombin Deficiency | Factor V Deficiency | Combined Factor V and Factor VIII Deficiency | Factor VII Deficiency | Factor X Deficiency | Factor XI Deficiency | Factor XIII Deficiency | Multiple Vitamin K–Dependent Factor Deficiency |
|---|---|---|---|---|---|---|---|---|---|
| | Less Common:<br>- Skin<br>- GI<br>- Genitourinary tract<br>- CNS | Less common:<br>- Menorrhagia<br>- CNS<br>- GI | Less common:<br>- Hematomas<br>- Hemarthrosis | - Postcircumcision<br>- Posttrauma<br>- Menorrhagia<br>- Postpartum | Less common:<br>- Hemarthrosis<br>- Hematoma<br>- Hematuria<br>- CNS<br>- GI | - After surgery<br>Less common:<br>- GI<br>- Umbilical cord bleeding<br>- Hematuria<br>- CNS | | - Mouth<br>- Gum bleeding<br>- Intracranial<br>Less common:<br>- Wound healing<br>- Hemarthrosis<br>- Epistaxis<br>- After surgery | - GI<br>- Easy bruising<br>- Mucocutaneous<br>- After surgery.<br>Children may show skeletal abnormalities. |
| **Risk of thrombosis** | **Afibrinogenemia**: reported.<br>**Dysfibrinogenemia**: reported. | In the inherited dysprothrombinemia caused by a G20210A mutation and linked to slightly increased levels of circulating prothrombin, there is a significantly higher risk to develop thromboembolic diseases | — | — | Thrombotic episodes and particularly deep vein thrombosis have been reported in 3–4% of patients (FVII-deficient patients seem to be not protected for venous and arterial thrombosis) | — | Cases of myocardial infarction and venous thrombosis have been reported (idiopathic or after FXI infusion) | — | Although protein S and protein C levels are low in VKCFD, there are no reports of venous or arterial thrombosis. |
| **Treatment: Replacement therapy** | - FFP<br>- Cryoprecipitate<br>- Fibrinogen plasma derived concentrate<br>- Recombinant human fibrinogen (rhFIB) has recently being developed and has received orphan drug designation from the U.S. FDA | - FFP<br>- PCC | - FFP<br>- Platelet transfusions (suggested in cases of severe bleeding not controlled with FFP or in case of inhibitor development) | **FV**: see FV column.<br>FVIII:<br>- FFP<br>- FVIII plasma derived concentrate<br>- Recombinant FVIII | - FFP<br>- FVII plasma derived concentrate<br>- Recombinant FVIIa | - FFP<br>- PCC | - FFP<br>- FXI plasma derived concentrate<br>- Recombinant FVIIa (few cases reported) | - FFP<br>- PCC<br>- FXIII plasma derived concentrate<br>- Recombinant FXIII (under study) | Oral or parenteral vitamin K. In patients who responded poorly to vitamin K and require surgical procedures or have acute bleeding, factor replacement is necessary with prothrombin complex. |
| **Treatment: Non-transfusional therapy (minor surgery, not severe bleeding, menorrhagia)** | - Estrogen-progestin preparations | - Antifibrinolytics<br>- Estrogen-progestin preparations | - Antifibrinolytic<br>- Levonorgestrel intrauterine<br>- Surgical treatments (menorrhagia) | - Antifibrinolytics<br>- Levonorgestrel intrauterine<br>- Desmopressin<br>- Estrogen-Progestin preparations | - Antifibrinolytics<br>- Estrogen-Progestin preparations | - Antifibrinolytics<br>- Estrogen-progestin preparations<br>- Levonorgestrel intrauterine | - Antifibrinolytics | - Antifibrinolytics<br>- Estrogen-progestin preparations | |

*This table provides a comparative synopsis of some important features of each of the distinct RBDs. The inheritance pattern for all these RBDs is autosomal recessive. Due to the lack of definitive data, this table does not provide information on typical ranges for defining deficiency severity and on the minimum level of factors necessary for effective hemostasis. Large data collections of patients affected with different RBDs is required to draw definitive conclusions regarding these issues.
†Data collected by World Federation of Hemophilia (WFH; www.wfh.org) and International Rare Bleeding Disorders Database (www.rbdd.org).
‡Ranges are approximate guides and will vary from laboratory to laboratory.
CNS, central nervous system; GI, gastrointestinal; ER, endoplasmic reticulum; FFP, fresh-frozen plasma; PCC, prothrombin complex concentrate; RBD, rare bleeding disorder.